



ArcGIS Pro SDK for .NET: Working with Desktop Application Markup Language (DAML)

Selim Dissem

2026 ESRI USER CONFERENCE

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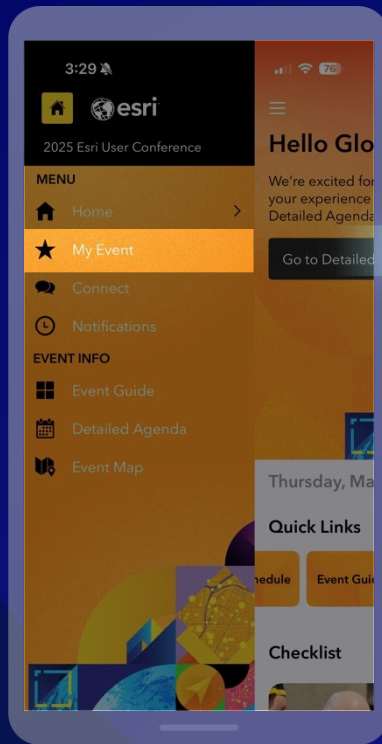
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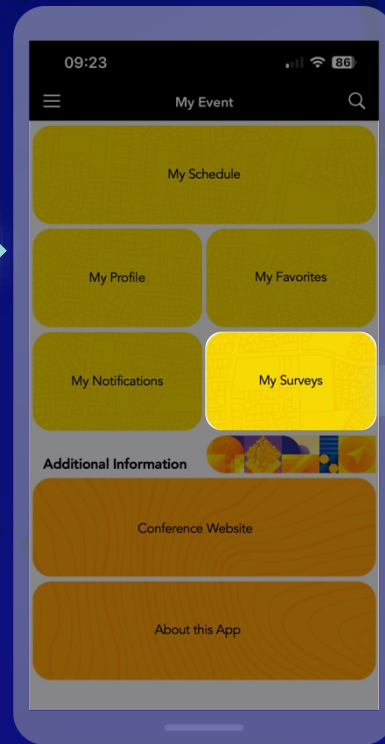
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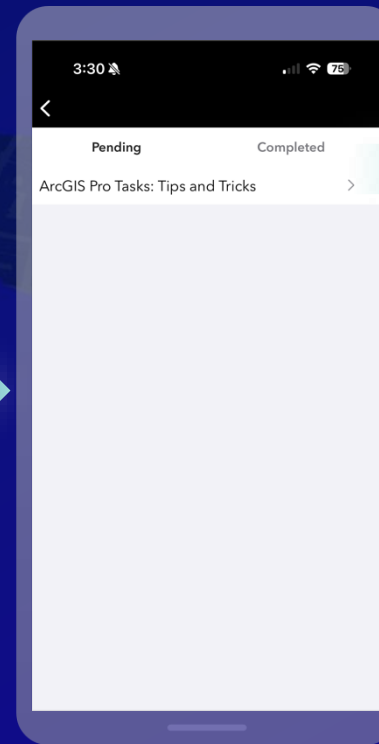
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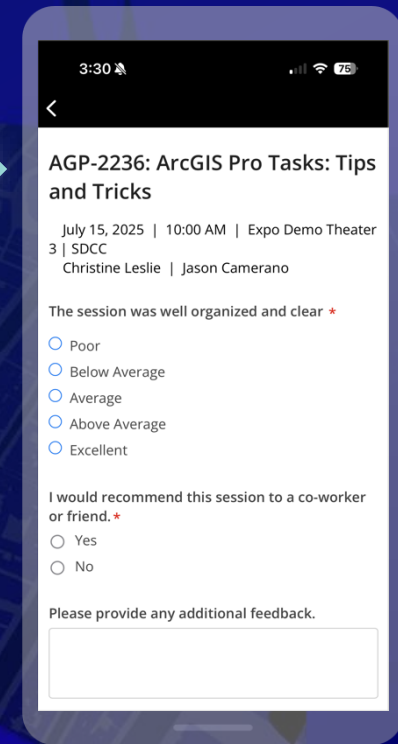
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Select the desired survey



Complete the survey



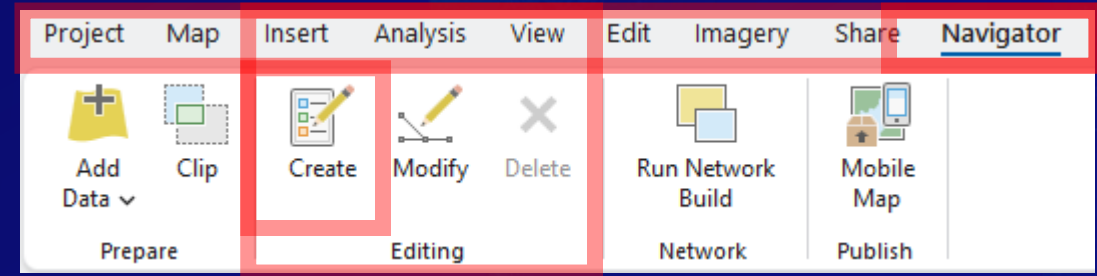
Session overview

- Introduction
 - Customize the ArcGIS Pro UI
 - What is DAML?
- Demos
 - Create a new add-in using an ArcGIS Pro Template
 - Create a Custom UI on the ArcGIS Pro ribbon
 - Reuse built-in ArcGIS Pro UI elements
 - Add a custom button
 - Enable / Disable the custom button contextually using States & Conditions
 - Modify the built-in ArcGIS Pro UI from my custom add-in
- Reference: “Introduction to DAML” in the ArcGIS Pro SDK Wiki



Introduction – Customize the ArcGIS Pro UI

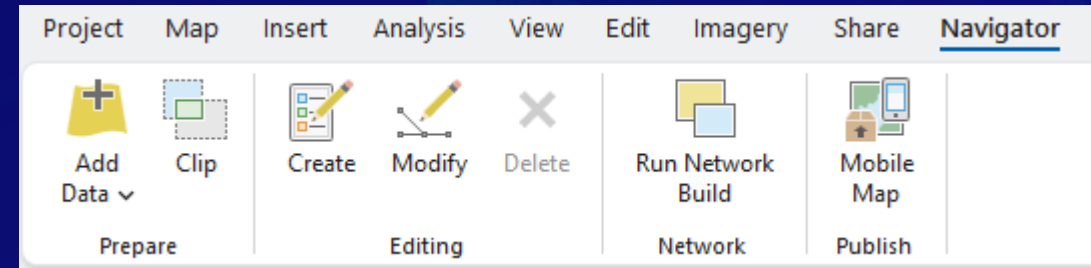
Customize the ArcGIS Pro UI



- ArcGIS Pro has a UI framework where each UI element is an extensibility point
- The UI elements of ArcGIS Pro that are most often customized are:
 - The ArcGIS Pro ribbon,
 - tabs,
 - groups,
 - other UI elements such as buttons, dockpanes, tools, galleries, backstage...
- To customize the ArcGIS Pro UI for our workflows we implement an ArcGIS Pro add-in
- ArcGIS Pro add-ins use Desktop Application Markup Language (DAML) to customize the ArcGIS Pro UI

What is Desktop Application Markup Language (DAML)?

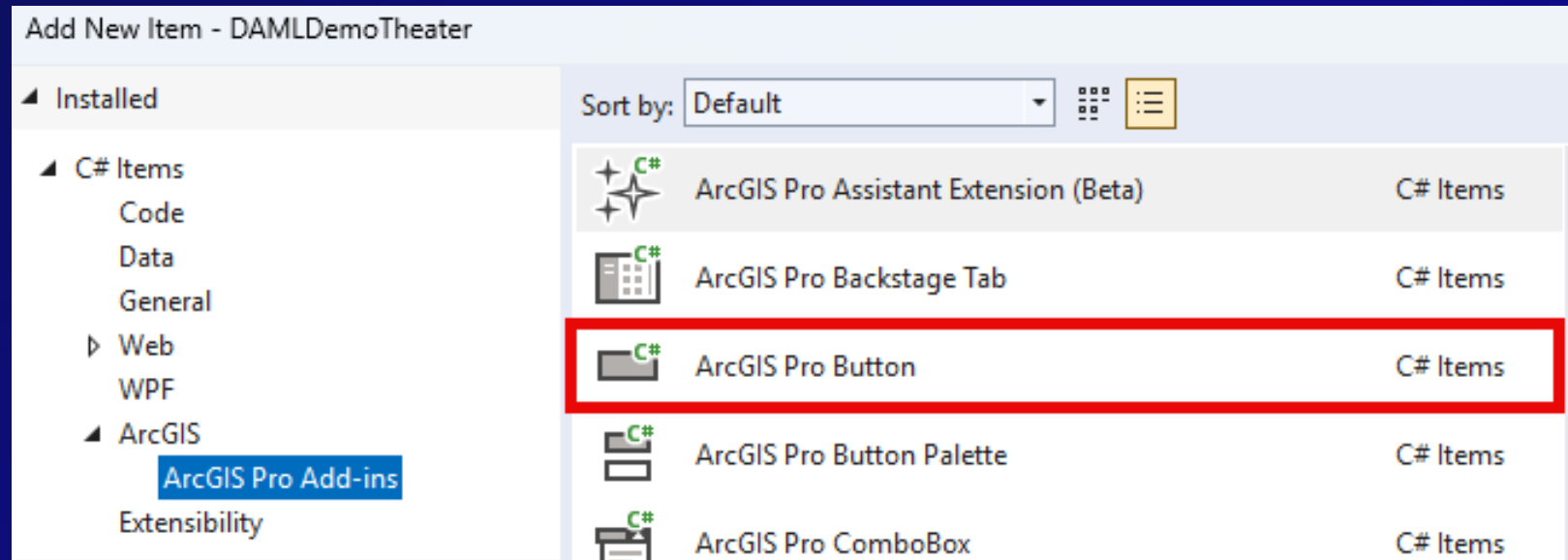
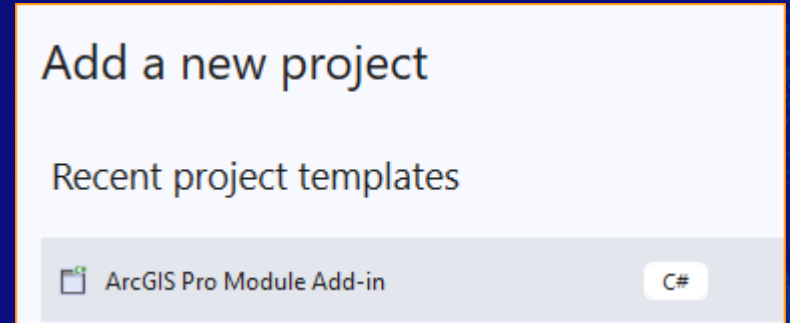
What is DAML?



- DAML uses XML Syntax, stored in a **Config.daml** file.
- All ArcGIS Pro Add-ins have a config.daml.
- DAML adds all new UI elements to the ArcGIS Pro UI.
- DAML can redefine and remove ArcGIS Pro UI elements.
- It controls UI contextual visibility through States and Conditions.
- The ArcGIS Pro SDK for .NET provides templates and samples to help developers create and manage these UI elements effectively.

Demo - Create a new add-in using an ArcGIS Pro Template

- In Visual Studio, use the ArcGIS Pro SDK module add-in project template
- Creates the config.daml required by the add-in
- Stubs out all basic classes required by the add-in
- Use ArcGIS Pro SDK Item templates to add supported UI elements in your add-in



Demo - Create a new add-in using an ArcGIS Pro Template

```
<?xml version="1.0" encoding="utf-8" ?>
<AddInManifest xmlns="http://schemas.esri.com/DADE/Registry"
  defaultAssembly="ProAppModule1.dll"
  defaultNamespace="ProAppModule1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://schemas.esri.com/DADE/Registry file:///C:/Program%20Files/ArcGIS/Pro/Extensions/Custom/ProAppModule1/Manifest.xml">
  <AddInInfo id="{8c872d29-78e3-4add-8031-57185f06110a}" version="1.0" desktopVersion="3.7.345">
    <Name>ProAppModule1</Name>
    <Description>ProAppModule1 description</Description>
    <Image>Images\AddinDesktop32.png</Image>
    <Author>seli2607</Author>
    <Company>ESRI</Company>
    <Date>2/20/2026 11:04:30 AM</Date>
    <Subject>Framework</Subject>
    <!-- Note subject can be one or more of these topics:
          Content, Framework, Editing, Geodatabase, Geometry, Geoprocessing, Layouts,
    -->
  </AddInInfo>
  <Modules>
    <insertModule id="ProAppModule1_Module" className="Module1" autoLoad="false" caption="Module1">
      <!-- uncomment to have the control hosted on a separate tab-->
      <tabs>
```

Demo context / requirements

- As an ArcGIS Navigator administrator, I'd like to have all the tools that I need to edit the data and publish it, all in one location.
- The tools identified for my work are:
 - Add data
 - Clip data
 - Create / Modify / Delete data
 - Build network based on the above data
 - Publish network as a mobile map



Data Management

What is Navigator?

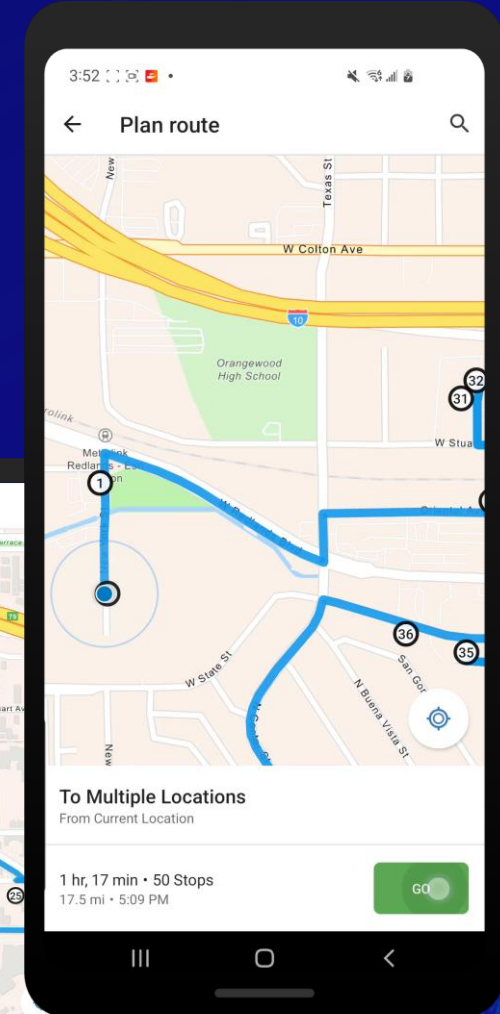
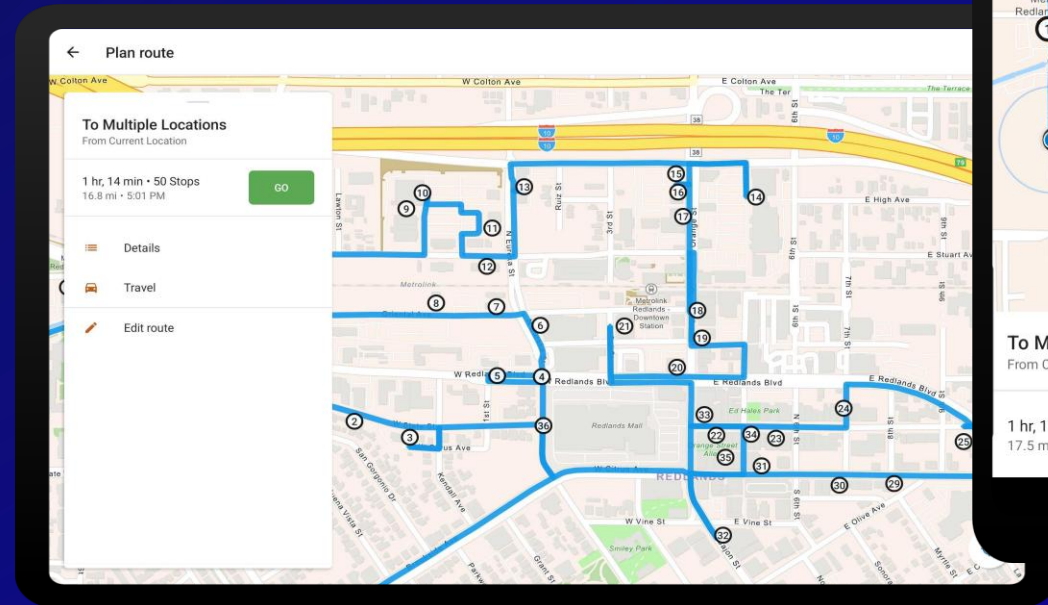
Advanced navigation and routing for your workforce

Custom Routing

Voice Guided, Turn-by-turn Navigation

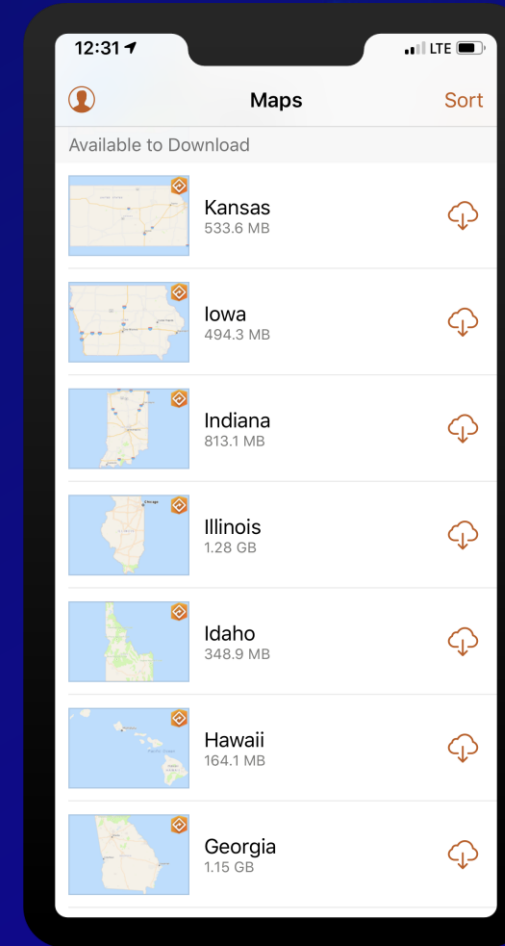
Navigation with your GIS Data

Reliable in Disconnected Environments



Customize Navigation

Routable networks that leverage your data

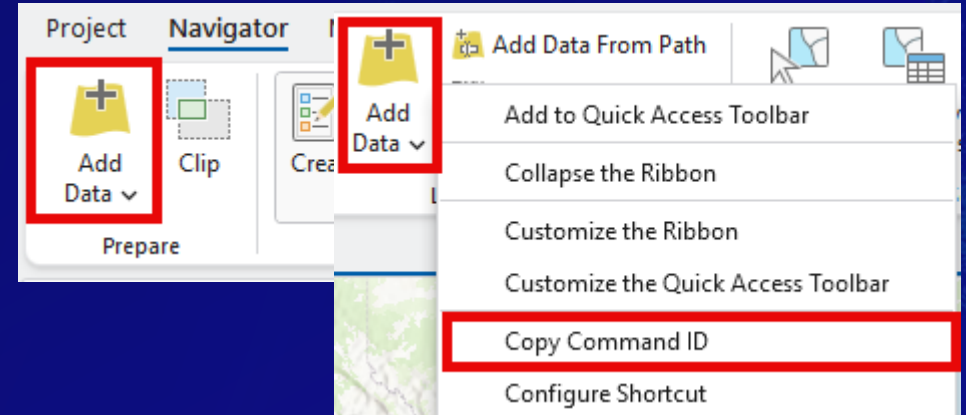


Routable networks,
operational layers,
locators, basemaps

ArcGIS Pro, Network Analyst, Streetmap Premium, Streetmap Premium Custom Roads

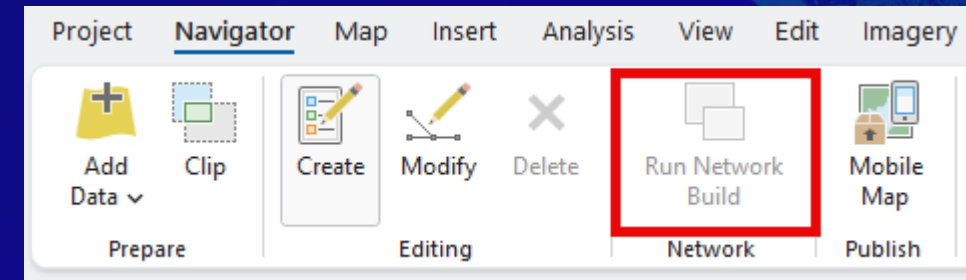
Demo - Create a Custom UI on the ArcGIS Pro ribbon

- Reuse built-in ArcGIS Pro UI elements
 - Each UI element has a Command ID
 - Use ArcGIS Pro options to find the needed Command ID
 - In your config.daml reference those IDs
- Add a custom button
 - Use ArcGIS Pro SDK Item templates to add our custom button
- Enable / Disable the custom button contextually using States & Conditions

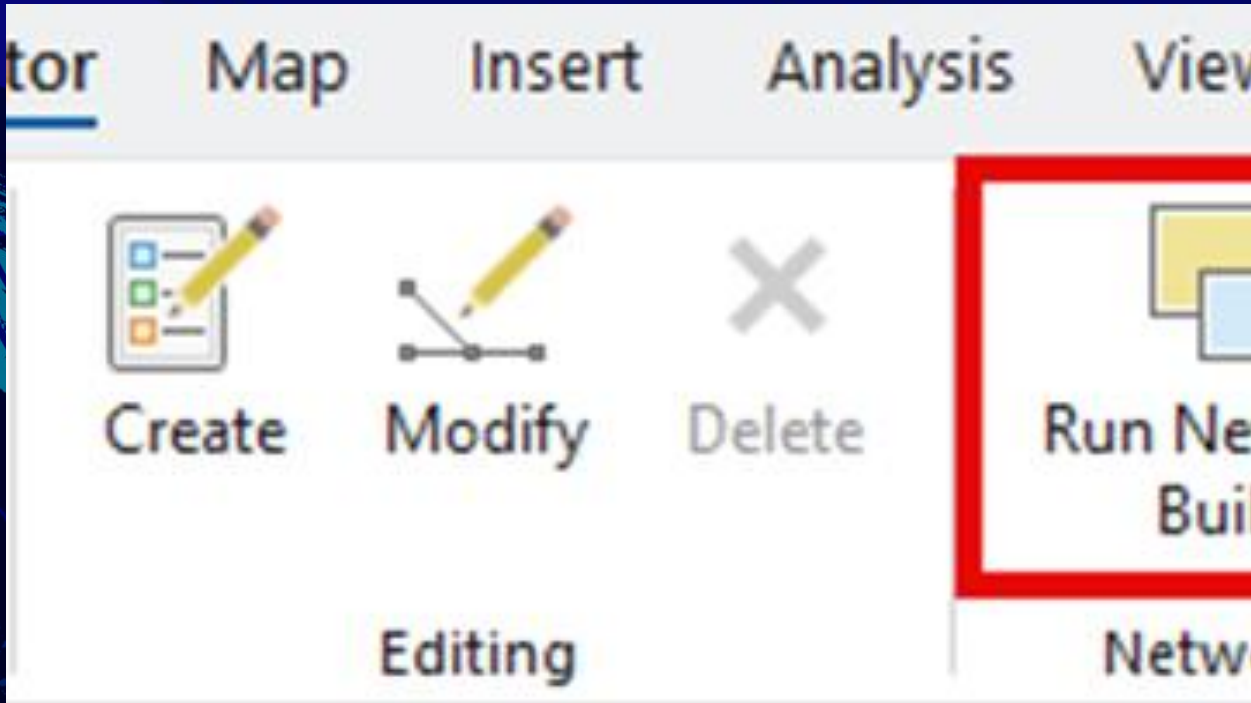


```
<conditions>
  <insertCondition id="NetworkDatasetLayerinMap_Condition">
    <state id="NetworkDatasetLayerinMap_State"/>
  </insertCondition>
</conditions>
```

```
<button id="AsyncGPBuildNetwork" caption="Run Network Build"
  condition="NetworkDatasetLayerinMap_Condition">
```

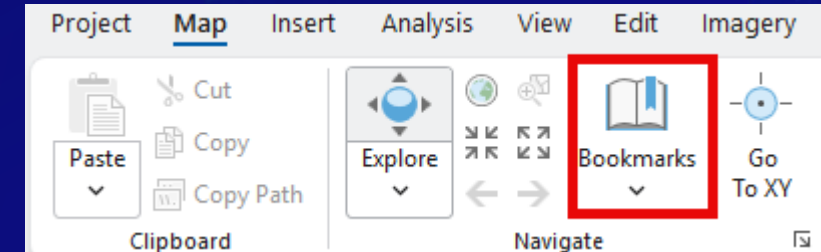


Demo - Create a Custom UI on the ArcGIS Pro ribbon

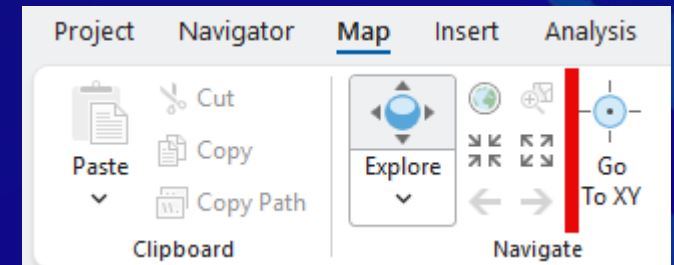


Demo - Modify the built-in ArcGIS Pro UI from my custom add-in

- Remove the Bookmarks button from the Navigate group
 - Find the Module ID (esri_mapping) and the Navigate group ID (esri_mapping_navigateGroup)
 - Find Command IDs through the respective extension DAML
 - Use Visual Studio Find in Files function to find esri_mapping_bookmarksNavigateGallery (Bookmarks Command ID)
 - Use DAML delete to remove it



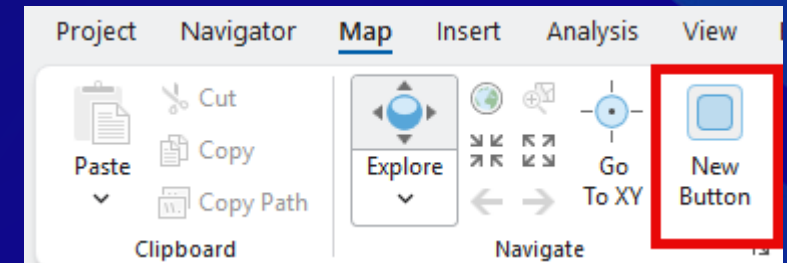
```
<updateModule refID="esri_mapping">
  <groups>
    <updateGroup refID="esri_mapping_navigateGroup">
      <deleteButton refID="esri_mapping_bookmarksNavigateGallery" />
    </updateGroup>
  </groups>
</updateModule>
```



Demo - Modify the built-in ArcGIS Pro UI from my custom add-in

- Add a new button to the Navigate group
 - Find the Module ID (esri_mapping) and the Navigate group ID (esri_mapping_navigateGroup)
 - Find Command IDs through the respective extension DAML
 - Use Visual Studio Find in Files function to find esri_mapping_bookmarksNavigateGallery (Bookmarks Command ID)
 - Use DAML insert to add it

```
<updateModule refID="esri_mapping">
  <groups>
    <updateGroup refID="esri_mapping_navigateGroup">
      <insertButton refID="New_Menu_Item_Button" />
    </updateGroup>
  </groups>
</updateModule>
```



Demo - Modify the built-in ArcGIS Pro UI from my custom add-in

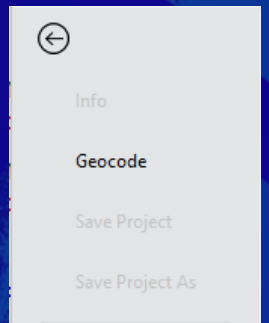
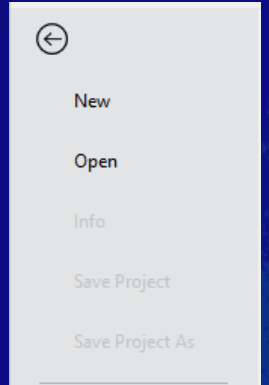
- Modify the backstage tabs

- Remove the New Project and Open Project tabs

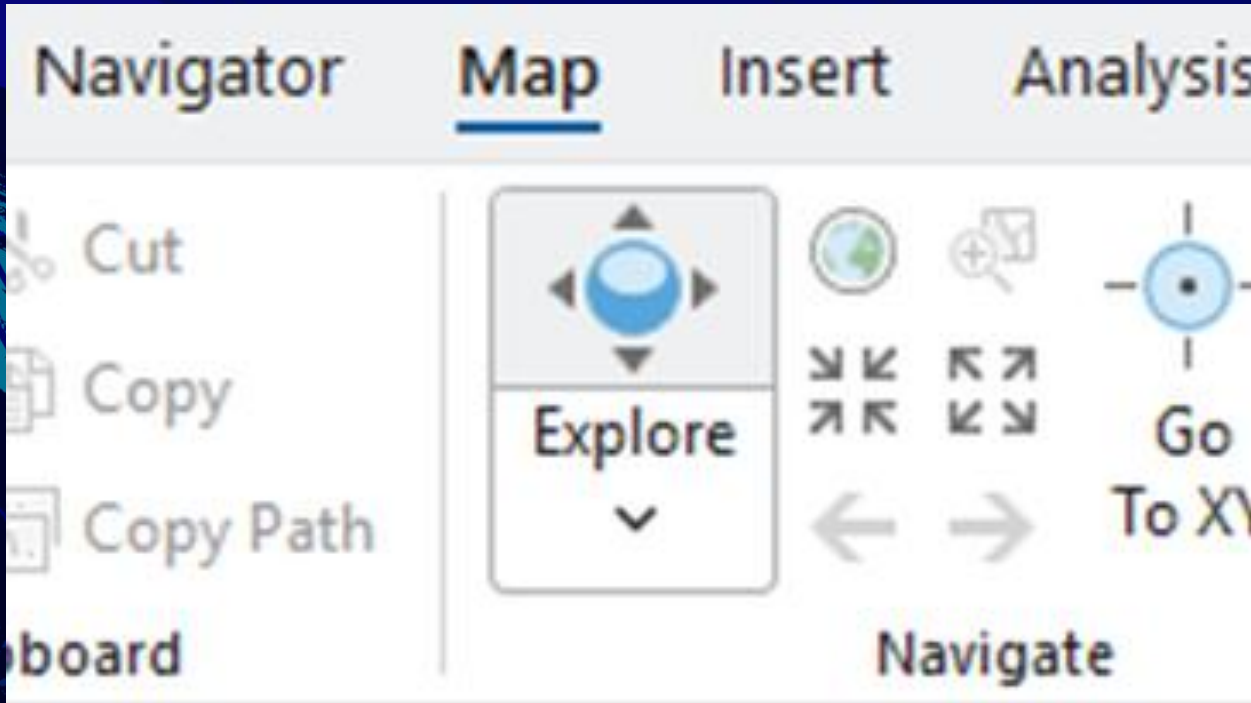
```
<!--Deletes the New Project tab in Pro's backstage-->  
<deleteTab refID="esri_core_newProjectBackStageTab" />  
<!--Deletes the Open Project tab in Pro's backstage-->  
<deleteTab refID="esri_core_openProjectBackStageTab" />
```

- Use DAML insert to add a new Geocode tab

```
<!--Inserts a new tab into Pro's backstage, placed before the Save Project tab-->  
<insertTab id="GeocodeBackstageTab" caption="Geocode"  
          className="DAML.Backstage.GeocodeBackstageTabViewModel"  
          insert="before" placeWith="esri_core_saveProjectButton" keytip="Z7">  
  <content className="DAML.Backstage.GeocodeBackstageTabView" />
```



Demo - Modify the built-in ArcGIS Pro UI from my custom add-in



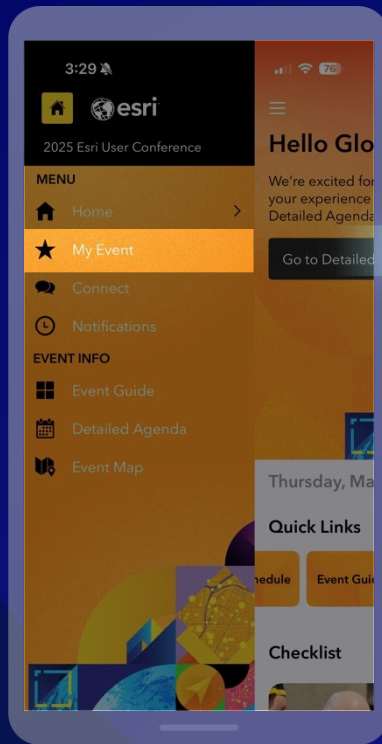
References

- <https://github.com/Esri/arcgis-pro-sdk/wiki/ProConcepts-Framework#introduction-to-daml-desktop-application-markup-language>
- <https://github.com/Esri/arcgis-pro-sdk/wiki/ProSnippets-DAML>
- <https://github.com/Esri/arcgis-pro-sdk-snippets/tree/main/SDKExamples/Framework/ProSnippet>
- <https://github.com/Esri/arcgis-pro-sdk-snippets/tree/main/SDKExamples/Framework/ProSnippet/DAML>
- <https://github.com/Esri/arcgis-pro-sdk/tree/master/DAML>
- <https://github.com/Esri/arcgis-pro-sdk-community-samples/tree/master/Framework/WorkingWithDAML>

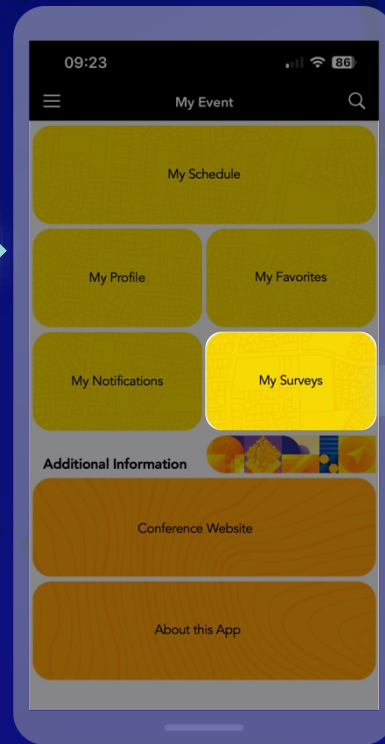


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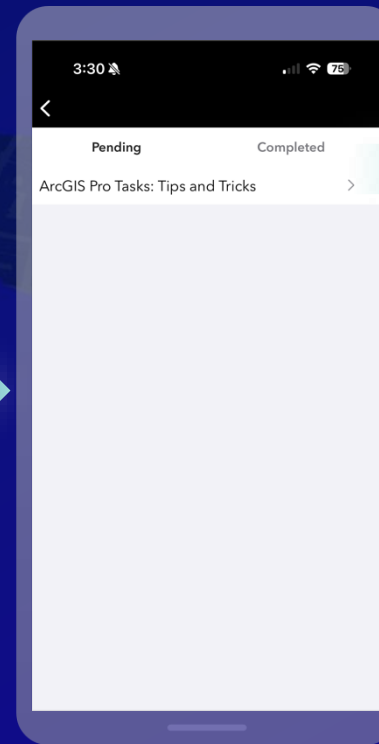
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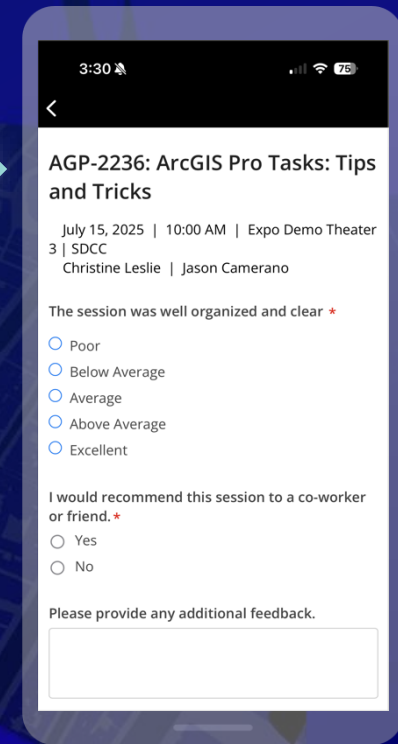
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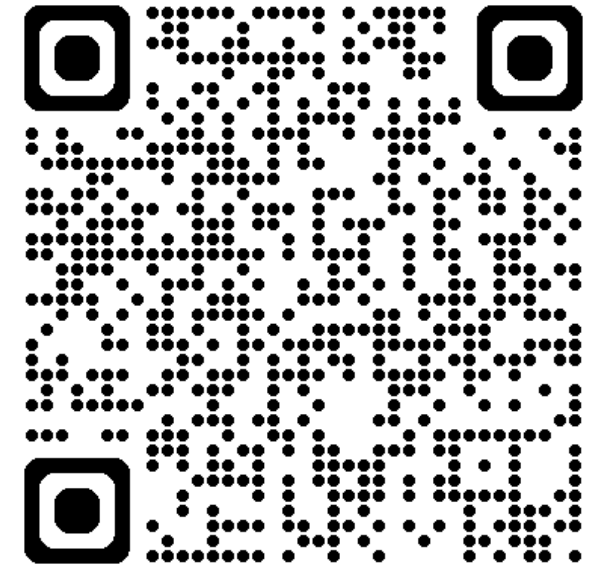


Complete the survey



Questions & Resources

- For questions, see us at the ArcGIS Pro SDK Showcase
- Session resources (code & slides) can be found here:
<https://github.com/EsriDevEvents/ArcGIS-Pro-SDK>



Questions

The background of the slide is split into two main sections. The upper section is a solid blue area featuring a white, intricate topographic map pattern of contour lines. The lower section is a curved, abstract shape with a gradient of colors ranging from deep purple to bright orange, resembling a stylized landscape or a celestial body like a sunset or a nebula.



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